

Units and Measurements

Quick reference — fold and keep.

MEASUREMENT IDENTITY

$$n_1 u_1 = n_2 u_2$$

SYSTEM CONVERSION

$$n_2 = n_1 [M_1/M_2]^a [L_1/L_2]^b [T_1/T_2]^c$$

FORCE / ENERGY (CGS)

$$1 \text{ N} = 10^5 \text{ dyne}, \quad 1 \text{ J} = 10^7 \text{ erg}$$

HOMOGENEITY

$$[\text{LHS}] = [\text{RHS}] = [\text{every term}]$$

RAYLEIGH METHOD

$$Q = k A^x B^y C^z$$

SIG FIGS: + AND -

fewest decimal places

SIG FIGS: × AND ÷

fewest significant figures

MEAN ABSOLUTE ERROR

$$\Delta a = \frac{1}{n} \sum |a_{\text{mean}} - a_i|$$

PERCENTAGE ERROR

$$\delta a\% = \frac{\Delta a}{a} \times 100$$

ERRORS: SUM / DIFFERENCE

$$\Delta Z = \Delta A + \Delta B$$

ERRORS: PRODUCT / QUOTIENT

$$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

ERRORS: POWERS

$$Z = A^p B^q C^{-r} \Rightarrow \frac{\Delta Z}{Z} = |p| \frac{\Delta A}{A} + |q| \frac{\Delta B}{B} + |r| \frac{\Delta C}{C}$$

VERNIER L.C.

$$\text{L.C.} = 1\text{MSD} - 1\text{VSD} = \frac{\text{MSD}}{n}$$

SCREW GAUGE L.C.

$$\text{L.C.} = \frac{\text{pitch}}{N}$$

ZERO CORRECTION

$$R_{\text{true}} = R_{\text{obs}} - \text{zero error}$$

DIMENSIONS: VISCOSITY

$$[\eta] = [\text{ML}^{-1}\text{T}^{-1}]$$

DIMENSIONS: PLANCK

$$[h] = [\text{ML}^2\text{T}^{-1}]$$

DIMENSIONS: G

$$[G] = [\text{M}^{-1}\text{L}^3\text{T}^{-2}]$$

Units and Measurements

Name: _____

Date: _____

Time allowed: 18 min

Marking: +4 for a correct answer, -1 for an incorrect multiple-choice answer, 0 for an unattempted question and for incorrect numerical answers. Shade one bubble per question on the grid below.

1. In the equation $y = A \sin(\omega t)$, the quantity ωt must be: ☒ G

(A) a length

(B) a time

(C) dimensionless

(D) a frequency

2. A screw gauge reads + 0.03 mm when its jaws are fully closed. The error this introduces is: ☒ G

(A) random

(B) systematic

(C) a gross error

(D) a least-count error

3. The dimensional formula of **acceleration** is: ☒ G

(A) $[LT^{-1}]$

(B) $[LT^{-2}]$

(C) $[MLT^{-2}]$

(D) $[L^2T^{-2}]$

4. The number of significant figures in 0.00450 is: ☒ G

(A) 2

(B) 3

(C) 5

(D) 6

5. A block has mass 4.237 g and volume 2.51 cm³. Report its density in g cm⁻³ to the correct number of significant figures. (M)

Answer: _____

6. The dimensional formula of **Planck's constant** h is: (M)

(A) [ML²T⁻¹]

(B) [ML²T⁻²]

(C) [MLT⁻¹]

(D) [ML⁻²T⁻¹]

7. In the equation $(P + \frac{a}{V^2})(V - b) = RT$, the dimensions of b are: (M)

(A) [ML⁵T⁻²]

(B) [L³]

(C) [ML⁻¹T⁻²]

(D) dimensionless

8. The speed of sound in a gas depends on pressure P and density ρ . By dimensional analysis, v is proportional to: (M)

(A) $P\rho$

(B) $\sqrt{P/\rho}$

(C) $\sqrt{P\rho}$

(D) P/ρ

9. In a system of units the unit of force is 100 N, the unit of length is 10 m and the unit of time is 100 s. The unit of mass in this system is 10 ^{n} kg. Find n . (M)

Answer: _____

10. A vernier calliper has L.C. = 0.01 cm. The main scale reads 1.2 cm and the 6th vernier division coincides. Give the reading in cm. (M)

Answer: _____

11. For the same data ($T_{\text{mean}} = 2.62$ s, $\Delta T = 0.11$ s), find the **percentage error**. (M)

Answer: _____

12. Five measurements of a time period give 2.63, 2.56, 2.42, 2.71, 2.80 s. Calculate the **mean absolute error** in seconds. (M)

Answer: _____

13. A calorie is 4.2 J. Suppose we adopt a system in which the unit of mass is α kg, of length β m and of time γ s. The new unit of energy is such that 1 calorie has magnitude: (H)

(A) $4.2 \alpha^{-1} \beta^{-2} \gamma^2$

(B) $4.2 \alpha \beta^2 \gamma^{-2}$

(C) $4.2 \alpha^{-1} \beta^2 \gamma^{-2}$

(D) $4.2 \alpha \beta^{-2} \gamma^2$

14. Luminous flux is measured in lumens, where $1 \text{ lm} = 1 \text{ cd} \cdot \text{sr}$. In terms of SI **base** units alone, the lumen is equivalent to: (H)

(A) cd m^2

(B) cd

(C) cd m^{-2}

(D) cd sr^2

15. A screw gauge has pitch 0.5 mm and 50 circular divisions. When the jaws are closed, the 45th division coincides with the reference line. A wire gives PSR = 1.5 mm and HSR = 27. Find the corrected diameter in mm. (H)

Answer: _____

Answer grid

1	☐ A	☐ B	☐ C	☐ D	2	☐ A	☐ B	☐ C	☐ D	3	☐ A	☐ B	☐ C	☐ D	4	☐ A	☐ B	☐ C	☐ D	5	_____
6	☐ A	☐ B	☐ C	☐ D	7	☐ A	☐ B	☐ C	☐ D	8	☐ A	☐ B	☐ C	☐ D	9	_____	10	_____	11	_____	
12	_____	13	☐ A	☐ B	☐ C	☐ D	14	☐ A	☐ B	☐ C	☐ D	15	_____								

Question paper

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Answer key

Keep this sheet face down until you have finished the paper.

Q	ANS	TIER	Q	ANS	TIER
1	C	G	9	5	M
2	B	G	10	1.26	M
3	B	G	11	4.2	M
4	B	G	12	0.11	M
5	1.69	M	13	A	H
6	A	M	14	B	H
7	B	M	15	1.82	H
8	B	M			

Worked solutions

1. Answer: C (G)

1. Trigonometric functions accept only pure numbers — a "sine of 3 metres" is meaningless.
2. Therefore $[\omega t] = [M^0 L^0 T^0]$.
3. It follows that $[\omega] = [T^{-1}]$, which is why ω is called angular *frequency*.

2. Answer: B (G)

1. Every single reading is shifted by the same + 0.03 mm, in the same direction.
2. Errors with a consistent sign and magnitude are **systematic**.
3. They cannot be averaged away — but they *can* be corrected by subtracting the zero error.

3. Answer: B (G)

1. Velocity has dimensions $[LT^{-1}]$.
2. Acceleration = $\frac{\Delta v}{\Delta t} = \frac{[LT^{-1}]}{[T]} = [LT^{-2}]$.
3. Note there is no M — acceleration does not care about mass.

4. Answer: B (G)

1. Leading zeros (0.00...) merely locate the decimal point — not significant.
2. That leaves 4, 5 and the trailing 0.
3. The trailing zero comes after a decimal point, so it **is** significant.
4. Answer: 3 significant figures. In scientific notation: 4.50×10^{-3} .

5. **Answer: 1.69** (M)

$$1. \rho = \frac{4.237}{2.51} = 1.68804...$$

2. 4.237 has 4 sig figs, 2.51 has 3. The answer keeps 3.

$$3. \rho = 1.69 \text{ g cm}^{-3}.$$

6. **Answer: A** (M)

1. From $E = h\nu$ we get $h = E/\nu$.

2. Energy: $[\text{ML}^2\text{T}^{-2}]$. Frequency: $[\text{T}^{-1}]$.

$$3. [h] = \frac{[\text{ML}^2\text{T}^{-2}]}{[\text{T}^{-1}]} = [\text{ML}^2\text{T}^{-1}].$$

4. Same as angular momentum — which is exactly why h shows up in the Bohr quantisation condition $L = nh/2\pi$.

7. **Answer: B** (M)

1. By the principle of homogeneity, b must have the same dimensions as V to be subtracted from it.

2. Volume has dimensions $[\text{L}^3]$.

3. So $[b] = [\text{L}^3]$ — physically, b is the excluded volume of the gas molecules themselves.

8. **Answer: B** (M)

1. Assume $v = kP^x\rho^y$, so $[\text{LT}^{-1}] = [\text{ML}^{-1}\text{T}^{-2}]^x[\text{ML}^{-3}]^y$.

2. From T: $-2x = -1 \Rightarrow x = \frac{1}{2}$. From M: $x + y = 0 \Rightarrow y = -\frac{1}{2}$.

3. Verify with L: $-x - 3y = -\frac{1}{2} + \frac{3}{2} = 1. \checkmark$

4. $v \propto \sqrt{P/\rho}$.

9. **Answer: 5** (M)

1. From $F = ma = \frac{mL}{T^2}$, we get $m = \frac{FT^2}{L}$.

2. Substituting the new units: $m = \frac{100 \times (100)^2}{10}$.

3. $= \frac{100 \times 10^4}{10} = 10^5 \text{ kg}$.

4. So $n = 5$.

10. **Answer: 1.26** (M)

1. Vernier contribution $= 6 \times 0.01 = 0.06 \text{ cm}$.

2. Reading $= 1.2 + 0.06 = 1.26 \text{ cm}$.

11. Answer: 4.2 (M)

1. Relative error = $\frac{\Delta T}{T_{\text{mean}}} = \frac{0.11}{2.62} = 0.0420$.
2. Percentage error = $0.0420 \times 100 = 4.2\%$.
3. The result is reported as $T = (2.62 \pm 0.11) \text{ s}$, or equivalently $2.62 \text{ s} \pm 4.2\%$.

12. Answer: 0.11 (M)

1. Mean = $\frac{13.12}{5} = 2.624 \approx 2.62 \text{ s}$.
2. Absolute deviations from the mean: $|2.63 - 2.62| = 0.01, 0.06, 0.20, 0.09, 0.18$.
3. Sum = 0.54.
4. Mean absolute error = $0.54/5 = 0.108 \approx 0.11 \text{ s}$.

13. Answer: A (H)

1. This is the classic NCERT/JEE problem. Energy is $[\text{ML}^2\text{T}^{-2}]$.
2. $n_2 = 4.2 \left[\frac{1 \text{ kg}}{\alpha \text{ kg}} \right]^1 \left[\frac{1 \text{ m}}{\beta \text{ m}} \right]^2 \left[\frac{1 \text{ s}}{\gamma \text{ s}} \right]^{-2}$
3. $= 4.2 \alpha^{-1} \beta^{-2} \gamma^2$.
4. The γ exponent flips sign because time appears with a negative power in the dimensional formula — this sign flip is exactly where most candidates lose the mark.

14. Answer: B (H)

1. The steradian is $\Omega = A/r^2$, an area divided by an area — completely dimensionless.
2. So multiplying by steradians changes nothing dimensionally.
3. $1 \text{ lm} = 1 \text{ cd} \cdot \text{sr} \equiv 1 \text{ cd}$ in base units.
4. The unit name *lumen* survives purely to tell a reader that flux, not intensity, is meant — the same trick as the radian.

15. Answer: 1.82 (H)

1. L.C. = $0.5/50 = 0.01 \text{ mm}$.
2. Closed reading: the 45th of 50 divisions coincides, which is 5 divisions *short* of zero.
3. Zero error = $-(50 - 45) \times 0.01 = -0.05 \text{ mm}$.
4. Observed = $1.5 + 27 \times 0.01 = 1.77 \text{ mm}$.
5. Corrected = $1.77 - (-0.05) = 1.82 \text{ mm}$.